

Data Structures & Algorithms

45-Days Curriculum

1. Summary

Phase	Title	Days
1	Python Programming	5
2	Data Structures & Algorithms (30 Problems)	40

2. Detailed Topic Outlines

Phase 1: Python Programming

Python Foundations

- Basics, variables, data types, strings, numbers, math methods.
- Conditional statements, loops, lambda, error handling.
- Functions, Built-In methods & higher-order functions.
- Variable scope, list, tuple, set, dictionary, arrays, objects, classes.

Phase 2: Data Structures & Algorithms (30 Problems)

Asymptotic Notations

- Time vs Space complexity.
- Best, average & worst case analysis.

Bit Manipulation

- AND, OR, XOR, NOT, Left/Right shift.
- Check Kth bit is set or not.

Recursion

- Introduction to recursion.
- Factorial, fibonacci, palindrome & subsets.

Arrays (17 Problems)

- Two Sum
- Best Time to Buy and Sell Stock
- Maximum Subarray
- Container With Most Water
- Trapping Rain Water
- Product of Array Except Self
- Maximum Product Subarray
- Maximum Sum Circular Subarray
- Majority Element
- Move Zeroes
- Sliding Window Maximum

- Subarray Sum Equals K
- Longest Subarray with Sum K (variant via LC patterns)
- Squares of a Sorted Array
- 3Sum
- Longest Consecutive Sequence
- Merge Intervals

Searching (3 Problems)

- Binary Search
- Search in Rotated Sorted Array
- Find Peak Element

Sorting (5 Problems)

- Bubble Sort
- Selection Sort
- Insertion Sort
- Merge Sort
- Quick Sort

Strings (5 Problems)

- Longest Substring Without Repeating Characters
- Longest Repeating Character Replacement
- Minimum Window Substring
- Valid Palindrome
- Find the Index of the First Occurrence in a String